



GAME DESIGN DOCUMENT (GDD) TEMPLATE

Game Name: Mars Colony

Genre: turn-based strategy

Game Elements:

Game elements are the basic activities the player will be doing for fun

- Worker placement
- Building placement
- Resource management

Player:

singleplayer



Civilization 6 – same genre game

TECHNICAL SPECS

Technical Form:

2D game on hexmap

View:

Game map will be observed from above (approximately 60 degrees)

Platform:

PC

Language:

C#

Device:

PC

GAME PLAY

turn 1

Player starts a game and is presented with a hex map full of empty sand places. They are also granted a greenhouse, excavator, personnel counting 10 people, batteries to supply electricity for greenhouse for 10 turns and food for 20 turns to feed 10 people. They place the greenhouse somewhere and assign 5 workers there (the more workers player assigns, the more productive it will be). Player then sends an excavator with the remaining people to discover what is under the sand right next to the greenhouse and ends turn.

turn 3

Excavator finds water source. Player assigns idle excavator with idle workers to search another place.

turn 6

Greenhouse finally produces some food. Food storage extends and player chooses what to do with idle workers.

Turn 10

Package from Earth arrives carrying batteries for 10 turns, 3 workers and blueprint for basic energy generator.

Game Play Outline

Opening the game application

As the loading screen, you see picture of decaying Earth with text over it. Text describes the problem with Earth and desperately asks player to solve the problem by colonizing Mars.



Dead space – illustration of decaying Earth

Game options

Simple graphics settings such as brightness, resolution.

Player can also choose difficulty by adjusting the natural disaster frequency and number of turns in which the colony must be sustainable.

Story synopsis

In the year 2100, Earth starts to be very polluted and running out of resources. People realize that Earth is unfit for living so they start to think of a solution to that problem. The solution seems to be to colonize Mars.

Modes

Only main campaign.

Game elements

Worker placement

Resource management

Colony expansion

Game levels

Early game – player is heavily relying on packages from Earth and starts to explore Mars.

Mid game – player tries to expand colony to most profitable locations and starts to make core infrastructure

End game – player can no longer count on support from Earth. Oxygen sources must be powered by minerals found and gathered here on Mars.

Player's controls

Player can do with only a mouse. There are several interactive buttons on the screen for example select idle workers, end turn, see available structures...

Winning

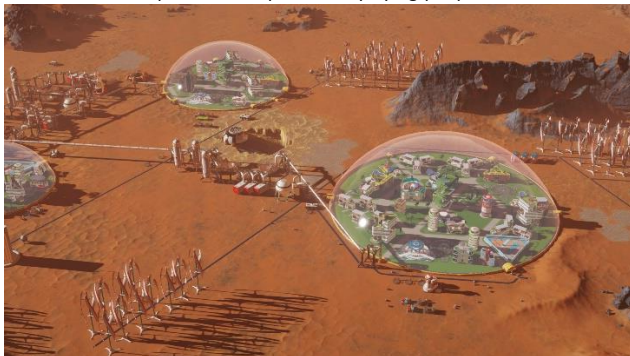
Being self-sustainable colony once the Earth stops supporting you.

Losing

When there are no more survivors on Mars.

End

Player gets either victory screen with picture of thriving colony and quotes from people thanking him or defeat screen showing the desolation of Earth and last quotes of desperation by dying people.



Surviving Mars – illustration of victory screen



Civilization 6 – loss screen

Why is all this fun?

Player must constantly think about his next move and strategize few turns ahead. Every decision can have consequences in the long run. Realistic scenario with high stakes makes player motivated to continue playing.

Key Features

Evolving Atmosphere

Turn-based strategy where every decision counts

DESIGN DOCUMENT

This document describes how GameObjects behave, how they're controlled and their properties. This is often referred to as the "mechanics" of the game. This documentation is primarily concerned with the game itself. This part of the document is meant to be modular, meaning you could have several different Game Design Documents attached to the Concept Document.

Design Guidelines

This is an important statement about any creative restrictions that need to be considered and includes brief statements about the general (i.e., overall) goal of the design.

- Art style and animations should be simplistic. We are working with a small team and the main focus should be on game mechanics and game balance.

Game Design Definitions

This section established the definition of the game play. Definitions should include how a player wins, loses, transitions between levels, and the main focus of the gameplay.

- After a time defined in the difficulty level, all support from earth stops coming and this puts players colony to a test. If the number of colonists doesn't drop below 10 before it stops deteriorating, player wins. In the game there are no levels and transition between early game, midgame and endgame are imaginary.

Game Flowchart

The game flowchart provides a visual of how the different game elements and their properties interact. Game flowcharts should represent Objects, Properties, and Actions present in the game. Each of these items should have a number reference to where they exist within the game mechanics document.

Menu

Simple game menu consists of 3 buttons (play, options, quit), when you hit play you can adjust difficulty and start a game or load last game from where it stopped. Background of the menu is a view of a desert from the surface of Mars.

Synopsis

Overall tone of the game is dark. The player is supposed to feel depressing atmosphere of the possible end of mankind.

Game Play

The campaign should be approximately 7 hours long consisting of turns which are short in the beginning (maximum of 3 minutes) but during midgame and endgame they will get longer and longer, because player has more difficult decisions to make and more workers to assign.



Civilization 5 – illustration of the overall tone of the game

Player Definitions

Actions

Build building, send excavator, assign workers, end turn

Player Properties

When player opens list of available structures and hovers his mouse over any of them, there should be a window showing resource yields from the building.

Whenever players assigns new workers to a building, there should be clearly visible window with yield changes with current workforce.

Player Rewards (power-ups and pick-ups)

The only thing that affects player without him working for the benefit is the supply package from Earth. This usually contains batteries, workers, food and some important materials for building infrastructure.

Other than that, every positive effect should be a result of players previous actions.

User Interface (UI)

UI in game should be very simplistic not to overwhelm player. There is a thin window on top of the screen which shows resources. Then there is a button on the left, which shows, how many workers are idle and by clicking the button, player can iterate over them. Right next to it, is a box that shows, when the next supply drop arrives and what will it bring. On the right, there is a button which ends current turn and the move progresses on to the next turn. Whenever player clicks on an empty hex on the map, there is a small popup window, which shows all possible things player can do on the hex such as send excavator here or list of structures that are available.

Workflow

My role in team: coder

Core mechanics:

[Art] – disaster animation (2)

[Sound] – early game music (7)

[Sound] – late game music optimistic (7)

[Sound] – late game music depressive (7)

[Sound] – disaster sound (2)

[Code] – win and loss trigger (1)

[Code] – disaster mechanics (2)

[Code] – difficulty control (1)

[Code] – logic behind oxygen consumption (2)

[Code] – game progression after every turn – yields and other random events, progress in construction (5)

Resource management:

[Art] – resource icons (1)

[Code] – make players bank of resources (1)

[Code] – worker assignment (2)

[Code] – decrease in food per worker (1)

[Code] – randomized packages from Earth (2)

[Code] – decrease in resources to provide power to oxygen generators (1)

Map:

[Art] – all different type of hexes (3)

[Code] – create map with interactive hexes (4)

[Code] – uncovering empty sand hexes (3)

[Code] – building placement (1)

Game objects:

[Art] – buildings + excavator (3)

[Art] – workers animation (1)

[Art] – ships from Earth (1)

[Sound] – short sound effect when clicking different objects in game (2)

[Code] – building prefabs with their yields (3)

UI:

[Art] – in game UI (2)

[Sound] – menu music (5)

[Code] – UI buttons (3)

Game options:

[Art] – Main Menu (1)

[Code] – load (1)

[Code] – save (1)

Final touch:

[Code] – testing balance of the game and adjusting stats accordingly (15)

I would say that it would be a good idea to have some time overhead so I would say that the minimum viable prototype should be done in three months.